

Customer Churn Prediction Project

Project Overview:

This project builds an end-to-end Machine Learning model to predict customer churn. It uses the provided Telco Customer Churn dataset.

Files Created:

1. churn_prediction.ipynb - A Jupyter notebook that loads the data, cleans it, preprocesses features, and trains three models: Logistic Regression, Random Forest, and XGBoost. It evaluates them using Accuracy, Precision, Recall, and F1-score, and saves the best model as 'best_model.joblib'.
2. app.py - A Streamlit user interface where users can input customer data and get real-time predictions on whether they will stay or leave.

How to Run:

1. Open 'churn_prediction.ipynb' and run all cells to train the model.
2. Open a terminal in this folder and run: 'streamlit run app.py'
3. A web browser will open with the interface.

Models Compared:

- Logistic Regression
- Random Forest Classifier
- XGBoost Classifier

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-Score